



EMOTION CLASSIFIER PROJECT

NATURAL LANGUAGE PROCESSING

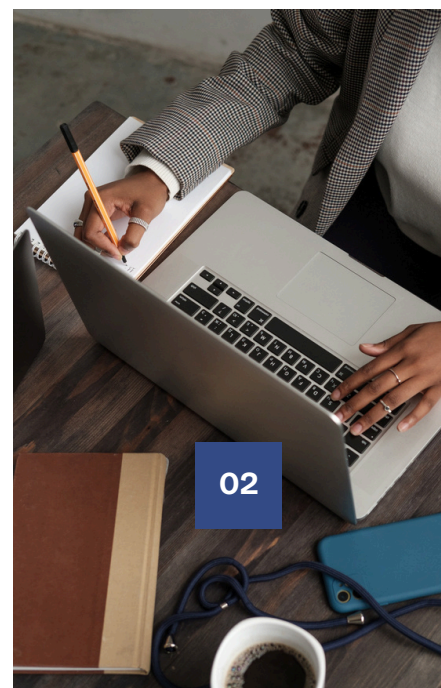
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INTRODUCTION

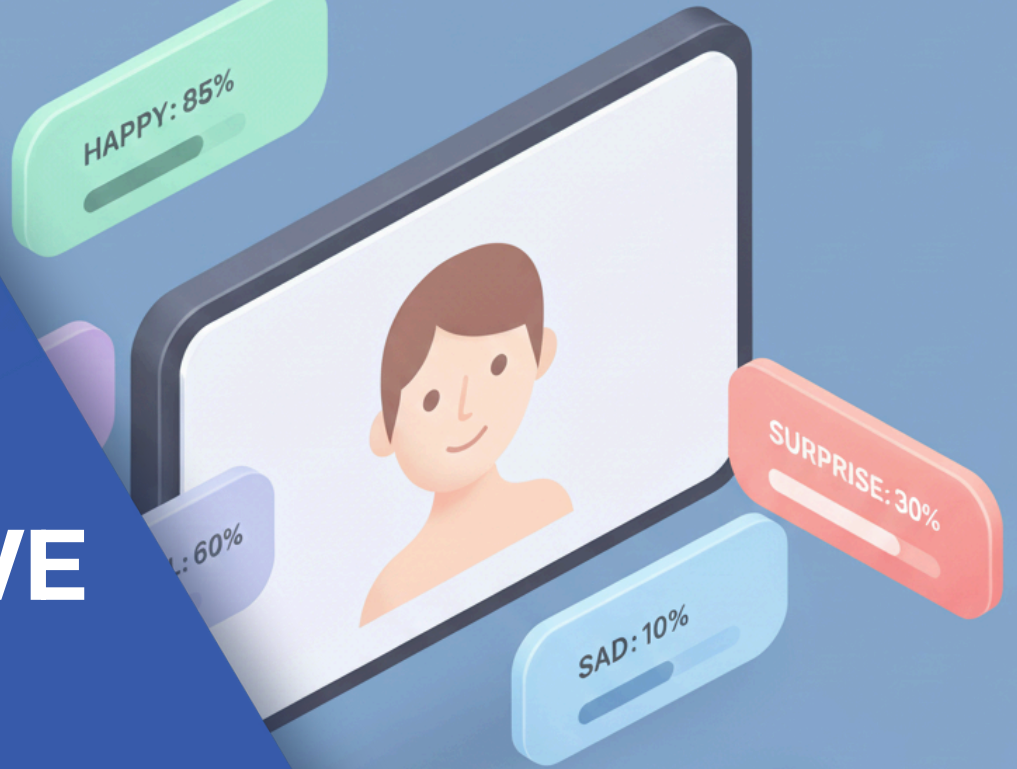
In today's digital world, people express their emotions largely through text on social media, chats, reviews, and online platforms. Understanding these emotions is essential for applications such as mental health analysis, customer feedback interpretation, and sentiment-aware systems. This project focuses on building an Emotion Classification system that automatically identifies the underlying emotion expressed in a given piece of text.

The model uses Natural Language Processing (NLP) techniques to convert textual data into meaningful numerical representations using TF-IDF vectorization. A Logistic Regression classifier is then trained on labeled emotion data to accurately categorize text into different emotional classes such as happiness, sadness, anger, fear, and others. The system is evaluated using standard performance metrics to ensure reliable predictions.

To make the solution practical and deployable, both the trained model and the text vectorizer are saved, enabling seamless integration into real-world applications such as web apps or APIs. This project demonstrates a complete machine learning pipeline—from data preprocessing and feature extraction to model training, evaluation, and deployment readiness—making it a strong foundation for advanced NLP-based emotion analysis systems.



OBJECTIVE



The primary objective of this project is to design and develop an automated emotion classification system that can accurately identify the emotional state expressed in a given text. By leveraging Natural Language Processing (NLP) and Machine Learning techniques, the project aims to convert unstructured textual data into meaningful insights related to human emotions.

- Build an emotion classification model using NLP
- Convert text into numerical features using TF-IDF
- Train a machine learning classifier
- Evaluate model performance
- Save the trained model for deployment

IMPORT REQUIRED LIBRARIES

Install Libraries

[1]:

```
import pandas as pd
import pickle

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report
```

DATASET IMPORT

Load Dataset

[2]:

```
df = pd.read_csv("emotions.csv")  
  
df.head()
```

[2]:

	text	label
0	i just feel really helpless and heavy hearted	4
1	ive enjoyed being able to slouch about relax a...	0
2	i gave up my internship with the dmrp and am f...	4
3	i dont know i feel so lost	0
4	i am a kindergarten teacher and i am thoroughl...	4

TRAIN -TEST SPLIT

[3]:

```
X = df["text"]  
y = df["label"]
```

Train-Test Split

[4]:

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y,  
    test_size=0.2,  
    random_state=42,  
    stratify=y  
)
```

TEXT FEATURE EXTRACTION

TF-IDF Vectorization

[5]:

```
vectorizer = TfidfVectorizer(  
    max_features=3000,  
    ngram_range=(1, 2),  
    stop_words="english"  
)  
  
X_train_tfidf = vectorizer.fit_transform(X_train)  
X_test_tfidf = vectorizer.transform(X_test)
```

TRAINING THE CLASSIFICATION MODEL

Train Logistic Regression

[6]:

```
model = LogisticRegression(  
    max_iter=1000,  
    class_weight="balanced",  
    n_jobs=-1  
)  
  
model.fit(X_train_tfidf, y_train)
```

[6]:

▼ LogisticRegression ⓘ ⓘ

► Parameters

EVALUATION

Evaluation

[7]:

```
y_pred = model.predict(X_test_tfidf)

accuracy = accuracy_score(y_test, y_pred)
print("Test Accuracy:", accuracy)
```

Test Accuracy: 0.9059763441376166



USER INTERFACE USING STREAMLIT

JOY

Emotion Classification App

Detect emotions from text using NLP

Enter your text:

This is the best day of my life

Predict Emotion

Predicted Emotion: JOY

Built with Streamlit, TF-IDF and Logistic Regression

SADNESS

Emotion Classification App

Detect emotions from text using NLP

Enter your text:

I feel so lonely and empty today.

Predict Emotion

Predicted Emotion: SADNESS

Built with Streamlit, TF-IDF and Logistic Regression

FEAR

Emotion Classification App

Detect emotions from text using NLP

Enter your text:

I am afraid of losing everything.

Predict Emotion

Predicted Emotion: FEAR

Built with Streamlit, TF-IDF and Logistic Regression

LOVE

Emotion Classification App

Detect emotions from text using NLP

Enter your text:

Being with you makes me feel loved and complete.

Predict Emotion

Predicted Emotion: LOVE

Built with Streamlit, TF-IDF and Logistic Regression

SURPRISE

Emotion Classification App

Detect emotions from text using NLP

Enter your text:

I am shocked by the sudden news

Predict Emotion

Predicted Emotion: SURPRISE

Built with Streamlit, TF-IDF and Logistic Regression



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